

MPM1D – Exam Review – Units 3-4 Homework

1. Gene works at a jean outlet store and she is paid \$23/hr. Let h be the number of hours worked and let E be her total earnings in dollars.

- a. Write an equation to represent Gene's earnings.

$$E = 23h$$

- b. Is this an example of direct variation or partial variation?

Direct. If she works 0 hours, she earns \$0

- c. What is the constant of variation?

23 - she earns \$23 each hour she works

- d. Construct a table of values to show Gene's earnings for 0, 1, 2, 3, 4, 5 hours.

h	E
0	0
1	23
2	46
3	69
4	92
5	115

- e. How much will Gene earn if she works for 8 hours?

$$\begin{aligned} E &= 23h \\ &= 23(8) \\ &= \$184 \end{aligned}$$

∴ She will earn \$184

2. The volume of a barrel is 22 litres but increases by 15 litres per minute someone pours water into it. Let V be total volume in litres and let t be time in minutes.

a. Write an equation to represent the volume of water in the barrel.

$$V = 15m + 22$$

let V be volume in L
" m " # of minutes H₂O is

b. Is this an example of direct variation or partial variation? paired in

Partial Variation b/c the bucket isn't empty to start

c. What is the constant of variation?

15 → the volume increases 15L each minute

d. What is the constant or fixed amount?

22 → the initial volume is 22L

e. What is the volume of water in the barrel after someone pouring in water for 8 minutes?

$$\begin{aligned} V &= 15m + 22 \\ &= 15(8) + 22 \\ &= 120 + 22 \\ &= 142 \text{ L} \end{aligned}$$

∴ The volume will be 142L

f. How long will it take for the barrel to have a volume of 217 litres?

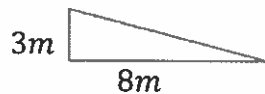
$$\begin{array}{rcl} V & = & 15m + 22 \\ 217 & = & 15m + 22 \\ -22 & & -22 \end{array}$$

$$\frac{195}{15} = \frac{15m}{15}$$

$$m = 13 \text{ min}$$

∴ It will take 13 minutes

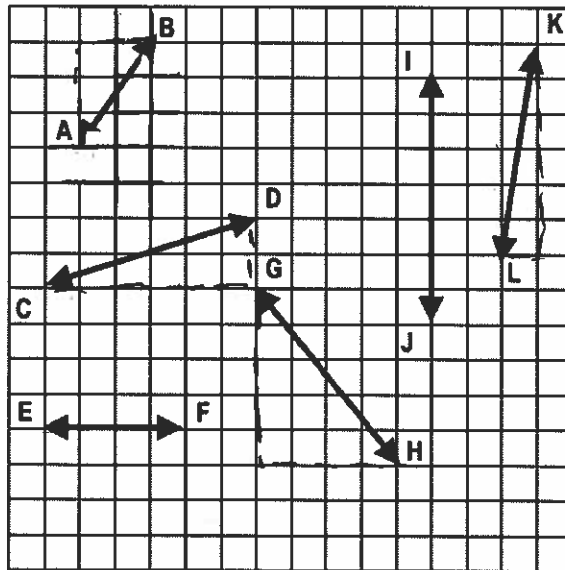
3. Determine the slope of the ramp?



$$m = \frac{\text{rise}}{\text{run}} = \frac{3}{8}$$

4. Determine the slope of each line. State the slope in lowest terms. Assume a one-unit grid.

Line	Slope
AB	$\frac{3}{2}$
CD	$\frac{1}{3}$
EF	0
GH	$-\frac{5}{4}$
IJ	undefined
KL	$\frac{6}{1} = 6$



$$\begin{aligned} CD: m &= \frac{\text{rise}}{\text{run}} \\ &= \frac{2}{6} \\ &= \frac{1}{3} \end{aligned}$$

$$\begin{aligned} EF: m &= \frac{\text{rise}}{\text{run}} \\ &= \frac{0}{4} \\ &= 0 \end{aligned}$$

$$\begin{aligned} IJ: m &= \frac{\text{rise}}{\text{run}} \\ &= \frac{-7}{0} \\ &= \text{undefined} \end{aligned}$$

5. Complete the table of values for the following linear relation and then write an equation to represent the linear relation.

	x	y	
	0	10	
+1 <	1	7	> -3
+1 <	2	4	> -3
+1 <	3	1	> -3
+1 <	4	-2	> -3
+1 <	5	-5	> -3

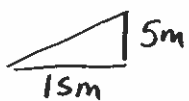
$$\begin{aligned} m &= \frac{\Delta y}{\Delta x} \\ &= \frac{-3}{1} \\ &= -3 \end{aligned}$$

$$b = 10$$

$$y = -3x + 10$$

6. A ramp rises 5 m over a run of 15 m.

- a. How would you change the run to make the slope steeper?



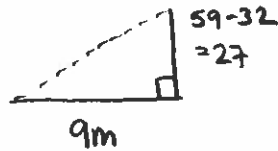
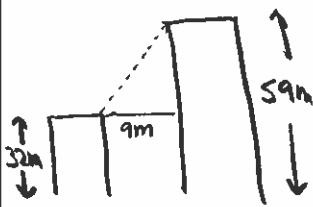
Decrease it

- b. How would you change the rise to make the slope steeper?

Increase it

$$\begin{aligned} m &= \frac{\text{rise}}{\text{run}} \\ &= \frac{5}{15} \end{aligned}$$

7. A steel beam goes between the tops of two buildings. The first building is 32 m high and the second building is 59 m high. If the two buildings are 9 m apart, what is the slope of the beam? Include a diagram in your solution.



$$m = \frac{\text{rise}}{\text{run}} \\ = \frac{27}{9} \\ = 3$$

8. Magic Mobile charges $\$37.50/\text{hr}$ to repair a cellphone in addition to an $\$85$ flat fee.

- a. Write an equation to represent the cost of getting a cellphone fixed at Magic Mobile. Let C be the cost in dollars and let h be the number of hours.

$$C = 37.50h + 85$$

- b. Is this an example of direct variation or partial variation? Explain.

Partial Variation b/c the cost doesn't start at 0. There is a fixed fee of $\$85$.

- c. How much would an individual have to pay if it took Magic Mobile three and one-half hours to repair their phone?

$$\begin{aligned} C &= 37.50h + 85 \\ &= (37.50)(3.5) + 85 \\ &= 131.25 + 85 \\ &= \$216.25 \end{aligned}$$

$\therefore$ It would cost $\$216.25$

9. Mr. T and Mrs. T are renewing their wedding vows. The venue for the reception charges a fee of $\$12000$ to rent the venue and $\$175$ per person.

- a. Write an equation to represent the cost of renting the venue for this celebration. Let C be cost in dollars and let n be the number of people that attend.

$$C = 175n + 12000$$

- b. Use your equation to determine how much will it cost if 200 people attend the reception.

$$\begin{aligned} C &= 175n + 12000 \\ &= 175(200) + 12000 \\ &= 35000 + 12000 \\ &= \$47,000 \end{aligned}$$

$\therefore$ It will cost
\$47000

- c. Use your equation to determine how many people attended the reception if the cost was \$67475.

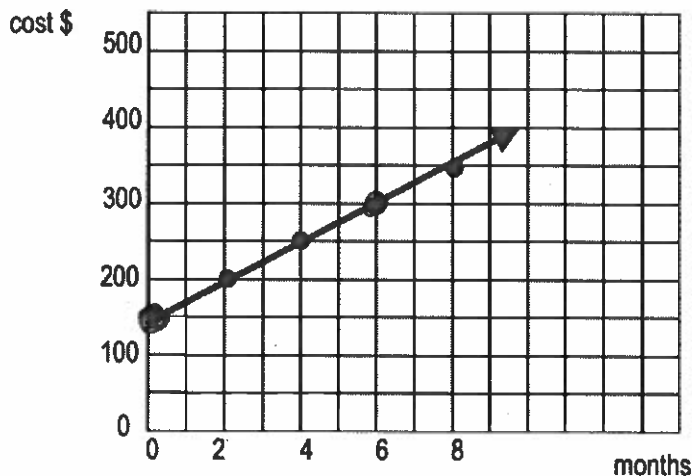
$$\begin{aligned} C &= 175n + 12000 \\ 67475 &= 175n + 12000 \\ -12000 & \qquad \qquad -12000 \end{aligned}$$

$$\frac{55475}{175} = \frac{175n}{175}$$

$$n = 317 \text{ people}$$

$\therefore$ 317 people attended

10. The graph below represents the cost to join a gym.



not a one-unit grid

no $\frac{\text{rise}}{\text{run}}$

a. Determine the slope of the line.

(0, 150)

(6, 300)

$$\begin{aligned}
 m &= \frac{y_2 - y_1}{x_2 - x_1} \\
 &= \frac{300 - 150}{6 - 0} \\
 &= \frac{150}{6} \\
 &= 25
 \end{aligned}$$

b. Explain the meaning of slope in this situation.

The gym costs \$25 per month

c. Determine the vertical intercept of the line.

$$b = 150$$

d. Explain the meaning of the vertical intercept in this situation.

The fixed amt to join the gym is \$150

e. Write an equation to represent the cost of being a member of this gym. Define your variables.

$$C = 25n + 150$$

let C be cost in \$
 " n " # of months

11. Using first differences, write an equation for each linear relation below.

	x	y	
+1 <	0	20	> +1
+1 <	1	21	> +2
+1 <	2	23	> +2
+1 <	3	25	> +2
+1 <	4	31	> +6

Not Linear

	x	y	
+3 <	-3	27	> +4
+3 <	0	31	> +4
+3 <	3	35	> +4
+3 <	6	39	> +4
+3 <	9	43	> +4

$$m = \frac{\Delta y}{\Delta x} \quad b = 31$$

$$= \frac{4}{3}$$

$$y = \frac{4}{3}x + 31$$

	x	y	
+1 <	-3	0	> +5
+1 <	-2	5	> +5
+1 <	-1	10	> +5
+1 <	0	15	> +5
+1 <	1	20	> +5

$$m = \frac{\Delta y}{\Delta x} \quad b = 15$$

$$= 5$$

$$y = 5x + 15$$

12. Use the RULE OF FOUR to represent the relation in three other ways. Assume the grid has a one-unit scale.

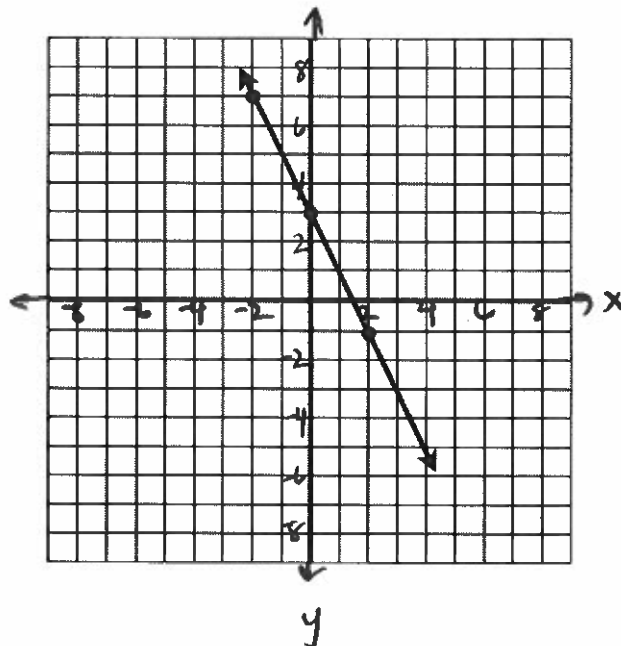


Table of Values

x	y
-2	7
0	3
2	-1

Eqⁿ

$$m = \frac{\text{rise}}{\text{run}}$$

$$= \frac{-4}{2}$$

$$= -2$$

$$b = 3$$

$$y = -2x + 3$$

Words

y ↓ by 2 when x ↑ by 1
y starts at 3

13. Identify the slope and y-intercept of each of the following linear equations.

Equation	Slope	Y-Intercept
$y = 3x + 5$	3	5
$y = -\frac{2}{3}x - 9$	$-\frac{2}{3}$	-9
$y = 6$	0	6
$x = -7$	undefined	non
$3x - 6y + 12 = 0$	$\frac{1}{2}$	2

horiz.

vert.

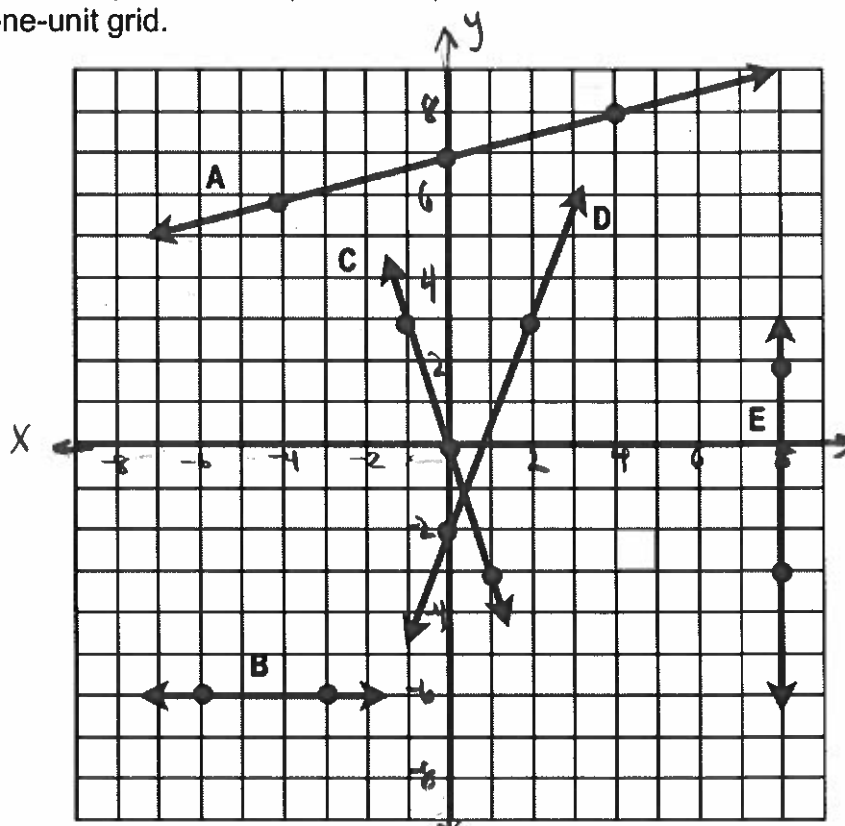
$$3x - 6y + 12 = 0$$

$\uparrow$ by $\uparrow$ by

$$\frac{3x}{6} + \frac{12}{6} = \frac{6y}{6}$$

$$y = \frac{1}{2}x + 2$$

14. Write an equation in slope-intercept form for each of the following lines. Assume a one-unit grid.



A: $m = \frac{\text{rise}}{\text{run}}$
 $= \frac{1}{4}$
 $b = 7$

$$y = \frac{1}{4}x + 7$$

B: $m = \frac{\text{rise}}{\text{run}}$
 $= \frac{0}{3}$
 $= 0$
 $b = -6$

$$y = -6$$

C: $m = \frac{\text{rise}}{\text{run}}$
 $= -\frac{3}{1}$
 $= -3$
 $b = 0$

$$y = -3x$$

D: $m = \frac{\text{rise}}{\text{run}}$
 $= \frac{5}{2}$
 $b = -2$

$$y = \frac{5}{2}x - 2$$

E: $m = \frac{\text{rise}}{\text{run}}$
 $= \frac{\text{undefined}}{0}$
 $= \text{undefined}$
 $x\text{-int} = 8$

$$x = 8$$

15. Graph each of the following equations on the grid provided.

a. $y = x + 4$

✓ b. $y = -\frac{3}{5}x - 2$

✓ c. $2x + 4y - 12 = 0$

✓ d. $y = -8$

✓ e. $x = 7$

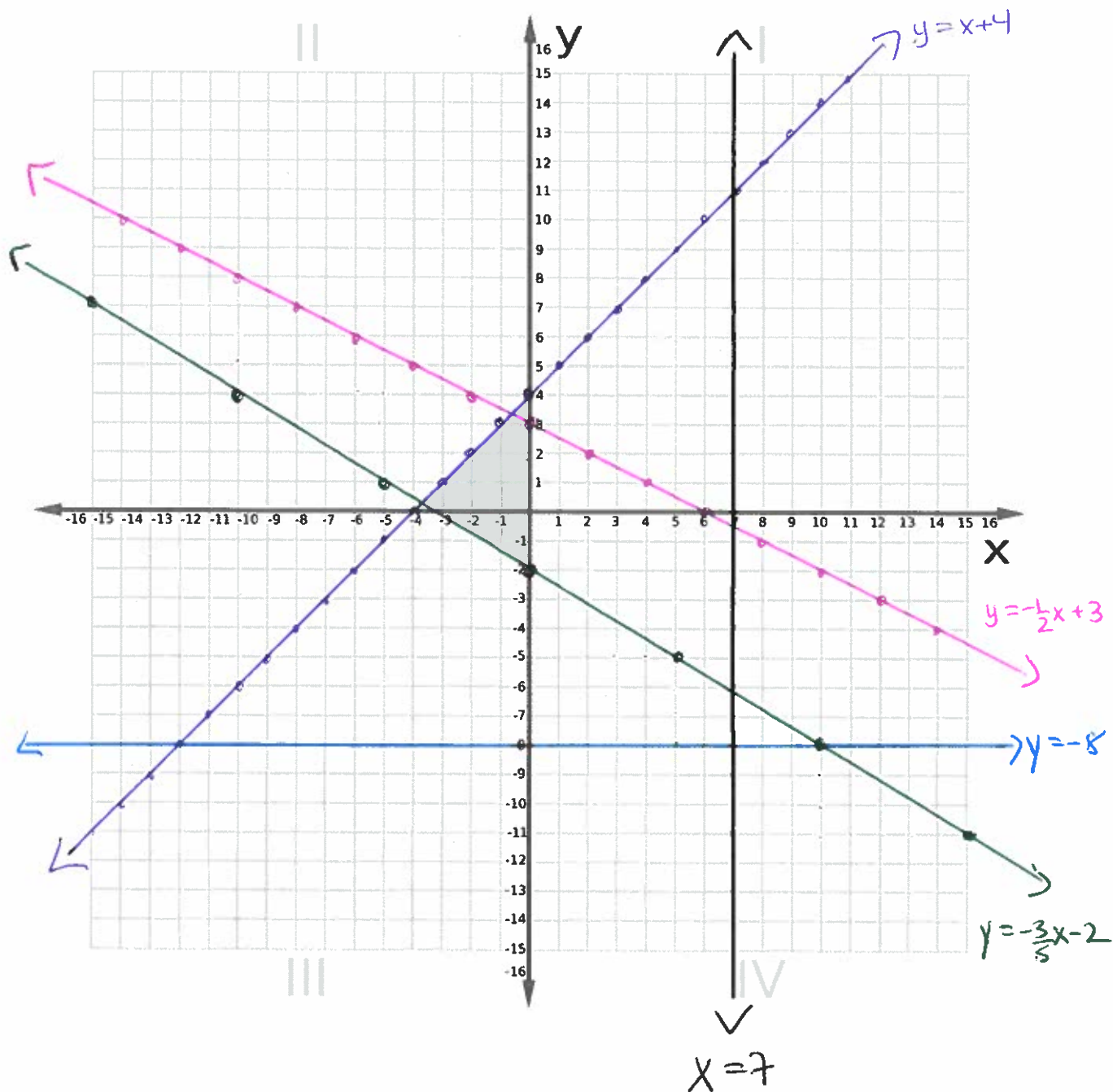
$$2x + 4y - 12 = 0$$

$$-4y = -2x + 12$$

$$y = \frac{1}{2}x - 3$$

$$\frac{2x}{-4} - \frac{12}{-4} = \frac{-4y}{-4}$$

$$y = -\frac{1}{2}x + 3$$



16. Rewrite each of the following equations. If the equation is in standard form, re-write it in slope-intercept form. If the equation is in slope-intercept form, re-write it in standard form.

a. $y = 3x - 8$

$$0 = 3x - y - 8$$

$$3x - y - 8 = 0$$

c. $3x - 6y - 12 = 0$

$$\frac{3x - 12}{+6} = \frac{+6y}{+6}$$

$$y = \frac{1}{2}x - 2$$

b. $y = -2x + 6$

$$0 = -2x - y + 6$$

$$0 = 2x + y - 6$$

$$2x + y - 6 = 0$$

d. $2x + 5y + 10 = 0$

$$\frac{2x + 10}{-5} = \frac{-5y}{-5}$$

$$y = -\frac{2}{5}x - 2$$

17. Determine the slope of the line passing through $(-2, 5)$ and $(8, -1)$. *only m needed

Pt 1 $(-2, 5)$

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

Pt 2 $(8, -1)$

$$= \frac{-1 - 5}{8 - (-2)}$$

$$= \frac{-6}{10}$$

$$= -\frac{3}{5}$$

$$y = mx + b$$

$$5 = \left(-\frac{3}{5}\right)\left(-\frac{2}{1}\right) + b$$

$$5^{(5)} = \frac{6}{5} + b^{(5)}$$

$$25_{-6} = 6 + 5b_{-6}$$

$$\frac{19}{5} = \frac{5b}{5} \quad b = \frac{19}{5}$$

$$\therefore y = -\frac{3}{5}x + \frac{19}{5}$$

18. Determine the x-intercept and y-intercept of the line $4x - 3y - 24 = 0$. Note the coordinates

X-int: let $y = 0$

$$4x - 3(0) - 24 = 0$$

$$4x - 24 = 0$$

$$\frac{4x}{4} = \frac{24}{4}$$

$$x = 6$$

Y-int: let $x = 0$

$$4(0) - 3y - 24 = 0$$

$$-3y - 24 = 0$$

$$-3y = 24$$

$$\frac{-3y}{-3} = \frac{24}{-3}$$

$$y = -8$$

or $4x - 3y - 24 = 0$

$$\frac{4x - 24}{3} = \frac{3y}{3}$$

$$y = \frac{4}{3}x - 8$$

$$b = -8$$

$\therefore$ The x-int is $(6, 0)$ & the y-int is $(0, -8)$

19. Write a linear equation in slope-intercept form for each of the following lines:

a. A line passing through the point $(3, -7)$ with a slope of 2.

$$m = 2$$

$$x = 3$$

$$y = -7$$

$$y = mx + b$$

$$-7 = (2)(3) + b$$

$$-7 = 6 + b$$

$$b = -13$$

$$\therefore y = 2x - 13$$

b. A line passing through the points $(-3, 8)$ and $(-7, 16)$.

pt 1 $(-7, 16)$

pt 2 $(-3, 8)$

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$= \frac{8 - 16}{-3 - (-7)}$$

$$= \frac{-8}{-3 + 7}$$

$$= \frac{-8}{4}$$

$$= -2$$

$$y = mx + b$$

$$8 = (-2)(-3) + b$$

$$8 = 6 + b$$

$$b = 2$$

$$y = -2x + 2$$

c. A line passing through $(-3, 8)$ with a slope perpendicular to the line

$$y = -\frac{1}{2}x + 9.$$

$$m = -\frac{1}{2}$$

$$|m_1 m_2| = 2$$

$$x = -3$$

$$y = 8$$

$$y = mx + b$$

$$8 = (-2)(-3) + b$$

$$8 = -6 + b$$

$$b = 14$$

$$\therefore y = 2x + 14$$

d. A line parallel to $2x + 4y - 9 = 0$ and with the same y-intercept as

$$9x + 6y - 36 = 0.$$

$$2x + 4y - 9 = 0$$

$$\frac{2x}{-4} - \frac{9}{-4} = \frac{-4y}{-4}$$

$$y = -\frac{1}{2}x + \frac{9}{4}$$

$$m = -\frac{1}{2} \quad |m_1| = -\frac{1}{2}$$

$$9x + 6y - 36 = 0$$

$$\frac{9x}{-6} - \frac{36}{-6} = \frac{-6y}{-6}$$

$$y = -\frac{3}{2}x + 6$$

$$|b| = 6$$

$$y = -\frac{1}{2}x + 6$$

20. A blue spruce tree is planted and grows at a constant rate each year. After 9 years, the tree is 458 cm in height. After 23 years the tree has a height of 976 cm.

a. Determine the constant rate that the tree grows each year.

(9, 458)

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

(23, 976)

$$= \frac{976 - 458}{23 - 9}$$

$$= \frac{518}{14}$$

$$= 37$$

$\therefore$ It grows 37 cm/yr

b. Determine the height of the tree when it was planted.

$$y = mx + b$$

$$458 = (37)(9) + b$$

$$458 = 333 + b$$

$$b = 125$$

$\therefore$ It was 125 cm tall when planted

c. Write an equation to represent the height of the tree given the number of years it has been growing

$$h = 37a + 125$$

Let h be the height in cm
" a " " age in yrs

d. What will the height of the tree be after 37 years?

$$h = 37a + 125$$

$$= 37(37) + 125$$

$$= 1369 + 125$$

$$= 1494 \text{ cm}$$

$\therefore$ It will be 1494 cm

Answer Key

- 1a) $E = 23h$ 1b) *Direct* 1c) 23 1d) See Solutions 1e) \$184
- 2a) $V = 15t + 22$ 2b) *Partial* 2c) 15 2d) 22
- 2e) 142 L 2f) 13 min
- 3) $m = \frac{3}{8}$
- 4) $AB = \frac{3}{2}$ $CD = \frac{1}{3}$ $EF = 0$ $GH = -\frac{5}{4}$ $IJ = \text{undefined}$ $KL = 6$
- 5) $y = -3x + 10$
- 6a) Decrease it 6b) Increase it
- 7) $m = 3$
- 8a) $C = 37.50h + 85$ 8b) *Partial* 8c) \$216.25
- 9) $C = 175n + 12000$ 9b) \$47000 9c) 317 people
- 10a) $m = 25$ 10b) Gym membership costs \$25 every month
- 10c) $b = 150$ 10d) The gym charges a fixed cost of \$150 to join
- 10e) $C = 25m + 150$ Let C be cost in \$ Let m be the number of months
- 11a) Not Linear 11b) $y = \frac{4}{3}x + 31$ 11c) $y = 5x + 15$
- 12) $y = -2x + 3$ Y increases by 2 when x increases by 1. Table: See solutions
Y starts at 3.
- 13) 3,5 13b) $-\frac{2}{3}, -9$ 13c) 0,6
- 13d) undefined, none 13e) $\frac{1}{2}, 2$
- 14) A: $y = \frac{1}{4}x + 7$ B: $y = -6$ C: $y = -3x$ D: $y = \frac{5}{2}x - 2$ E: $x = 8$
- 15) See solutions

16a) $3x - y - 8 = 0$ 16b) $2x + y - 6 = 0$ 16c) $y = \frac{1}{2}x - 2$ 16d) $y = -\frac{2}{5}x - 2$

17) $m = -\frac{3}{5}$

18)
 $(6, 0)$ and $(0, -8)$

19a) $y = 2x - 13$ 19b) $y = -2x + 2$ 19c) $y = 2x + 14$ 19d) $y = -\frac{1}{2}x + 6$

20a) 37 cm/yr 20b) 125 cm

20c) $H = 37t + 125$ Let H be height in cm. Let t be time growing in years

20d) 1494 cm